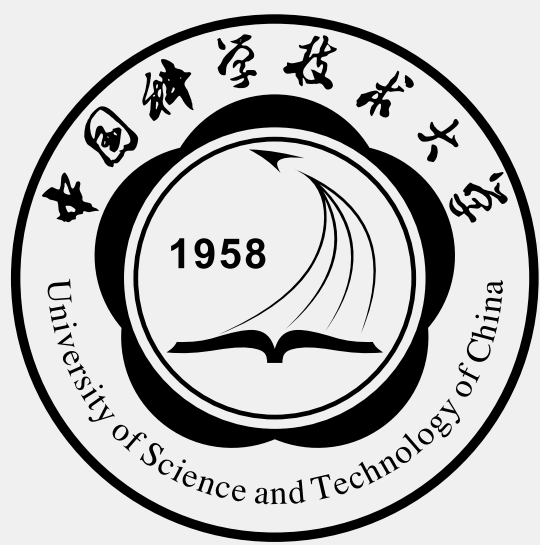




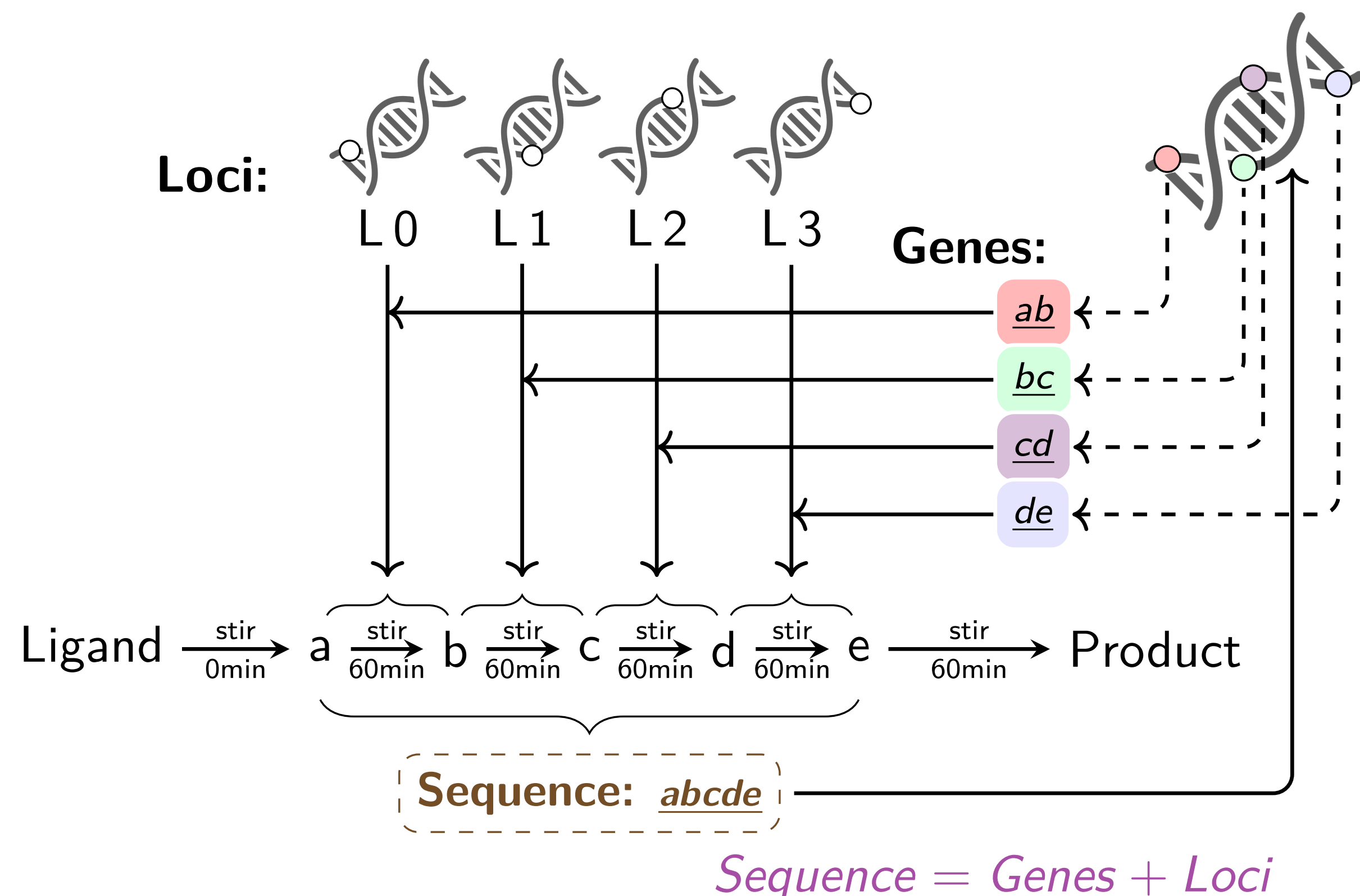
Algorithm-Driven Unraveling of Genetic Influence on High-Entropy Metal-Organic Catalysts

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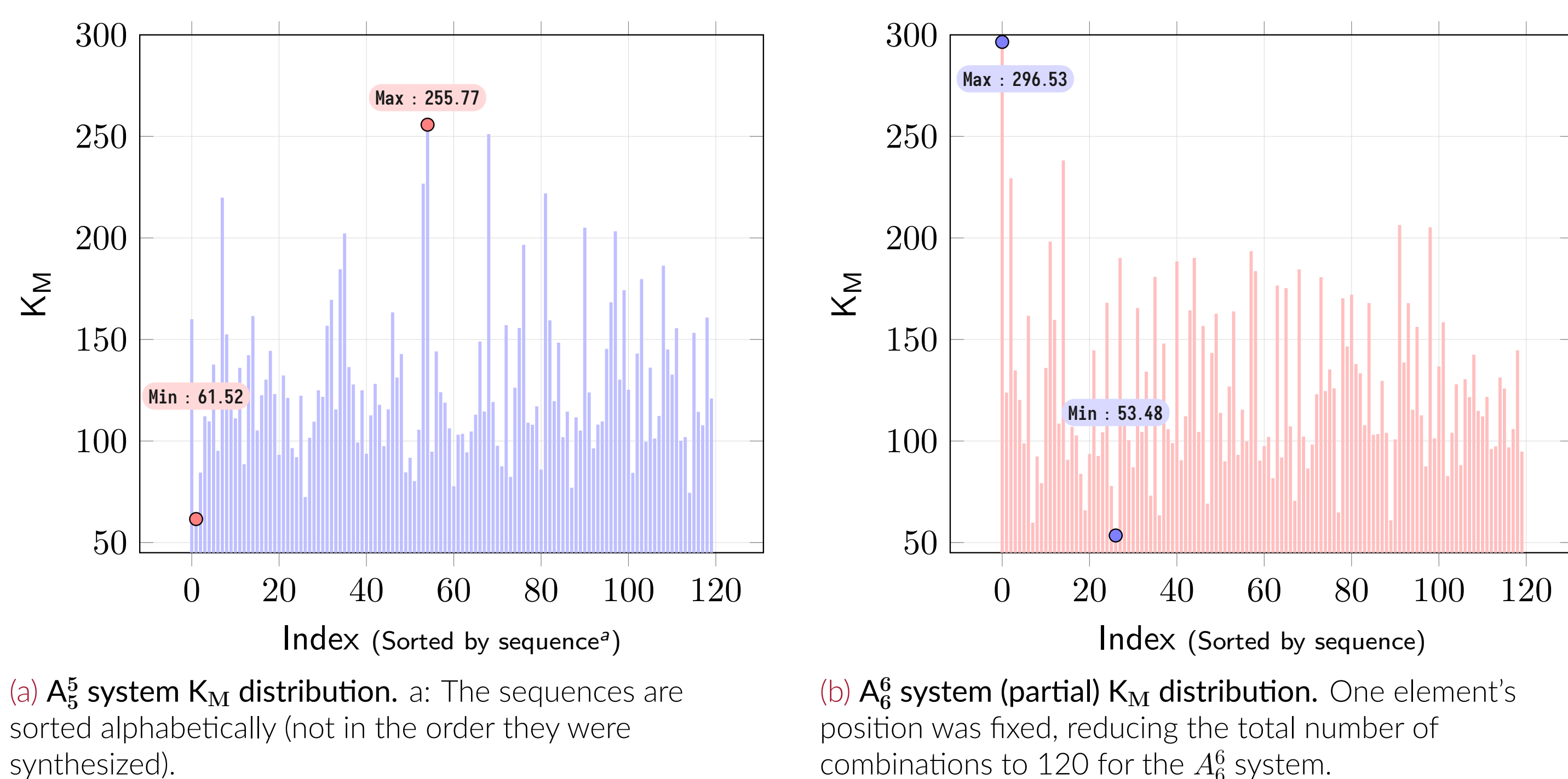
Abstract

- High-Entropy Materials:** *High-entropy materials (HEMs)* show promise for catalysis, but their complexity presents challenges for mechanistic insights and performance optimization.
- HEMOC and Temporal Sequence:** In our study on *High Entropy Metal Organic Catalysts (HEMOC)*, we found that the metal addition order, or *temporal sequence (TS)*, significantly affects *peroxidase-like (POD)* activity, with K_M values ranging from **255.8** to **61.5**.
- DNA-Inspired Model:** We represent metal addition sequences as “DNA” and adjacent metal pairs as **genes**, with sequence positions as **loci**.
- Gene Influence:** Our findings suggest that these **genes** and their **loci** influence HEMOC performance, validated by a custom algorithm achieving up to **1.80x** speedup over random optimization.
- Design Implications:** This work provides new insights into catalyst material development, reveals the underlying principles of the HEMOC system, and offers a new algorithm for optimized search within *temporal sequence (TS)* systems.



Experimental Observation: Over Fourfold Variation in K_M

- TS-HEMOC:** Adjusted metal addition order in synthesizing HEMOC.
- A_5^3 System:** Fully combinatorial sequence of five metallic acetates (Mn, Co, Ni, Cu, Zn).
- A_6^0 System (partial):** Partially constrained six-metal system (Mn, Co, Ni, Cu, Zn, Cd) with Co fixed in the third position, reducing complexity.



Descriptor Extraction: Genes and Loci

- Poor descriptor:** The position of individual elements (nodes), it does not clearly distinguish between high and low performance.
- Good descriptor:** Combinations of adjacent elements and their positions (lines between nodes).

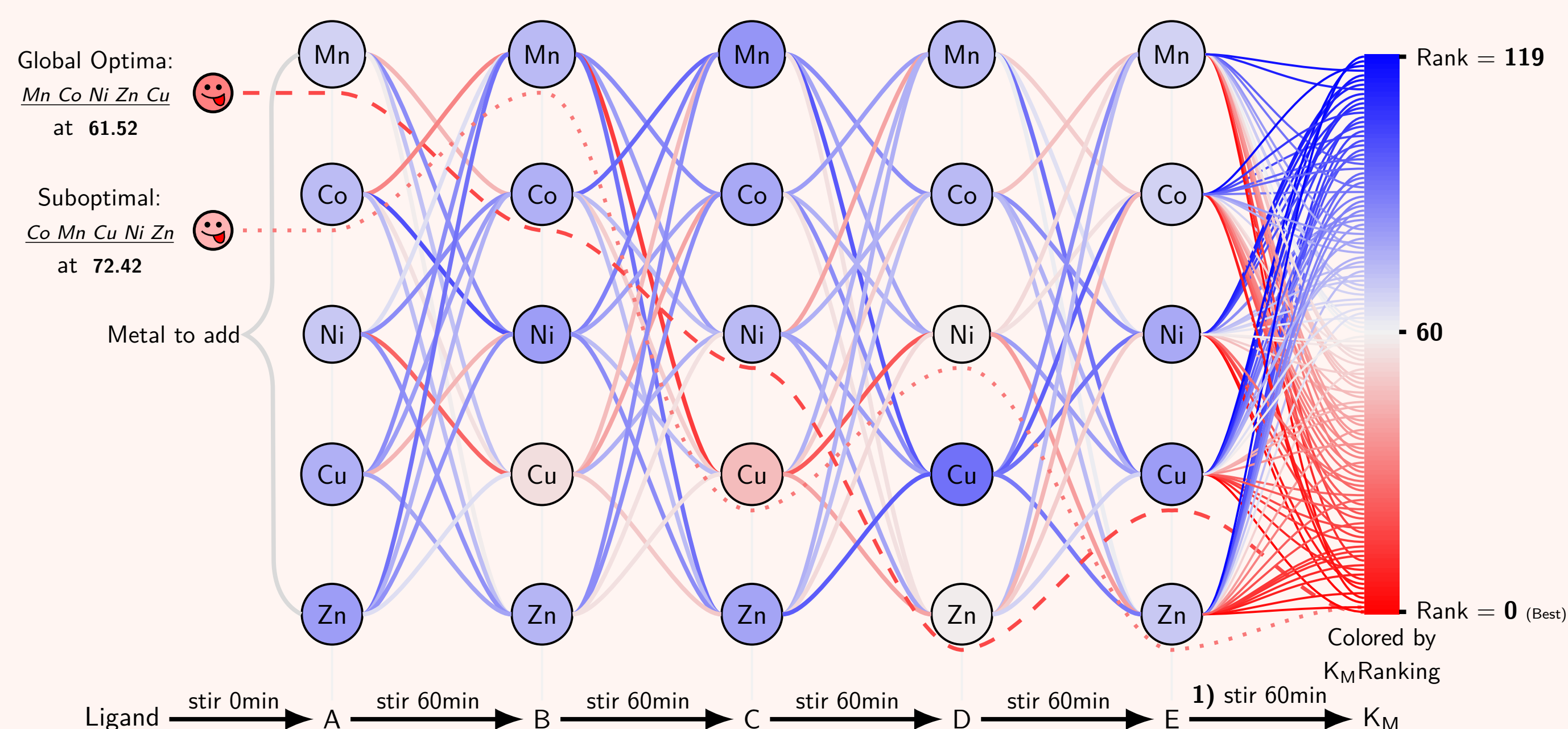


Figure 2. Distribution of temporal sequences in the A_5^3 system, illustrating the influence of metal pairs at different positions on K_M values. Each path from the left to right represents a metal addition sequence, with nodes indicating the metal added at each position (A to E). The lines connecting the nodes represent metal pairs, and their color indicates the catalytic performance: red lines correspond to higher-performing pairs, and blue lines to lower-performing ones. The shading of each line reflects the average K_M ranking for sequences that pass through those pairs at that specific position. The optimal and suboptimal sequences are highlighted on the left. This diagram demonstrates how the composition of metal pairs at each step significantly influences the overall catalytic activity.

Virtual Iterative Optimization: Validating the Gene-Locus Hypothesis

Inspired by Genes and Loci Hypothesis, we developed the **GLM Algorithm**.

- Input:** Known sequences and their K_M values.
- Output:** Predicted scores for unknown sequences.
- Principle:** Deconstructs high-dimensional sequences into adjacent metal pairs (**genes**) and their positions (**loci**), assuming neighboring elements have the strongest impact on catalytic performance, consistent with surface catalysis principles.
- GLM:** Stores scores for each gene at each locus by analyzing known K_M sequences, then uses these scores to recommend optimal unknown sequences.

The contenders are ready. Let's dive into an exciting **benchmark** showdown!

- Dataset:** A_5^3 and A_6^0 (partial) systems.
- Algorithms:** Our gene-locus-based **GLM Optimization Algorithm**, a baseline random search algorithm, and the **Genetic Algorithm (GA)**.
- Objective:** Benchmark and validate our hypothesis by demonstrating the superior performance of the GLM Optimization Algorithm (if it outperforms others).
- Iterations:** 1000 iterations (averaged).

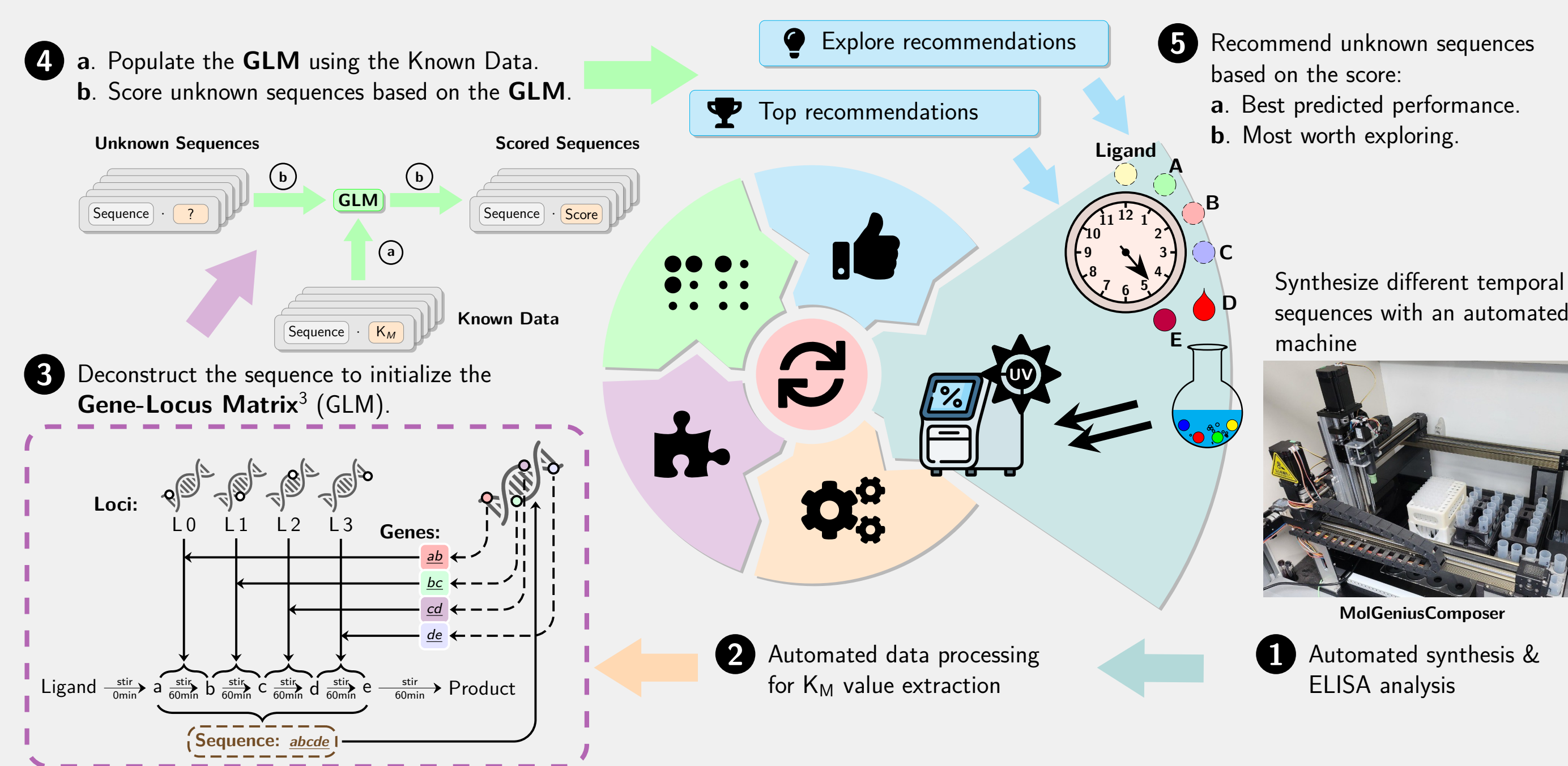


Figure 3. Iterative optimization cycle based on the gene-locus hypothesis. The cycle consists of five steps: (1) The automated system synthesizes unknown samples and performs ELISA analysis to measure catalytic performance. (2) The absorbance data is processed to fit the Michaelis-Menten equation, yielding the K_M value. (3) The *Gene-Locus Matrix (GLM)* is initialized with all possible gene-locus combinations. (4) Known sequences are used to populate the GLM, while unknown sequences are scored based on the attributes of their gene-locus pairs within the GLM. (5) The system recommends sequences with the best predicted performance or most worth exploring.

Benchmark Results: Outstanding Performance of the GLM Algorithm

- GLM Algorithm Performance:** Achieved a **1.80x** speedup over random search in the A_5^3 system, demonstrating effective sequence optimization.
- Partial A_6^0 System:** Speedup reduced to **1.38x** due to data constraints and fixed Co position, but still showed valuable insights.
- Validation of Gene-Locus Hypothesis:** Results support gene-locus interactions as effective descriptors for guiding optimization.

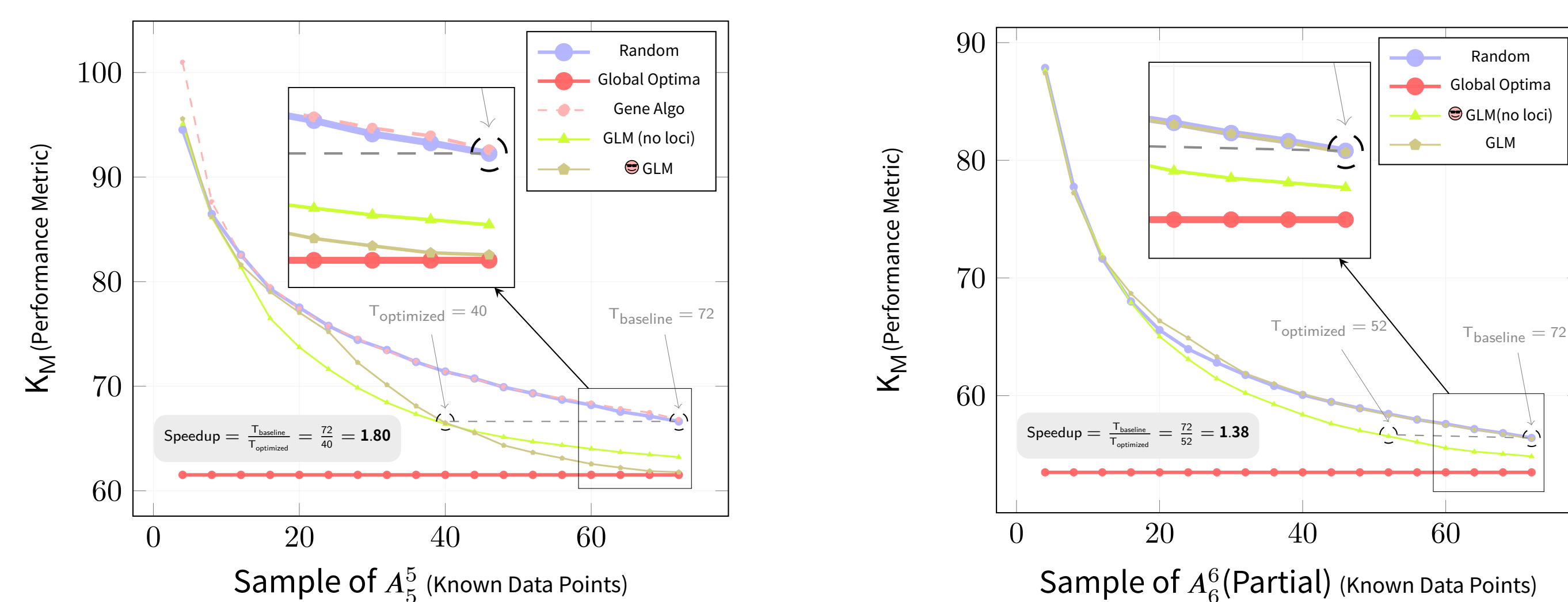


Figure 4. Optimization performance in the A_5^3 and A_6^0 (partial) systems. The vertical axis shows the best K_M value found (representing catalytic performance), while the horizontal axis indicates the number of groups (known data points) used in the optimization process. The results represent the average of **1,000** trials.

Method: Automated Mixing Device

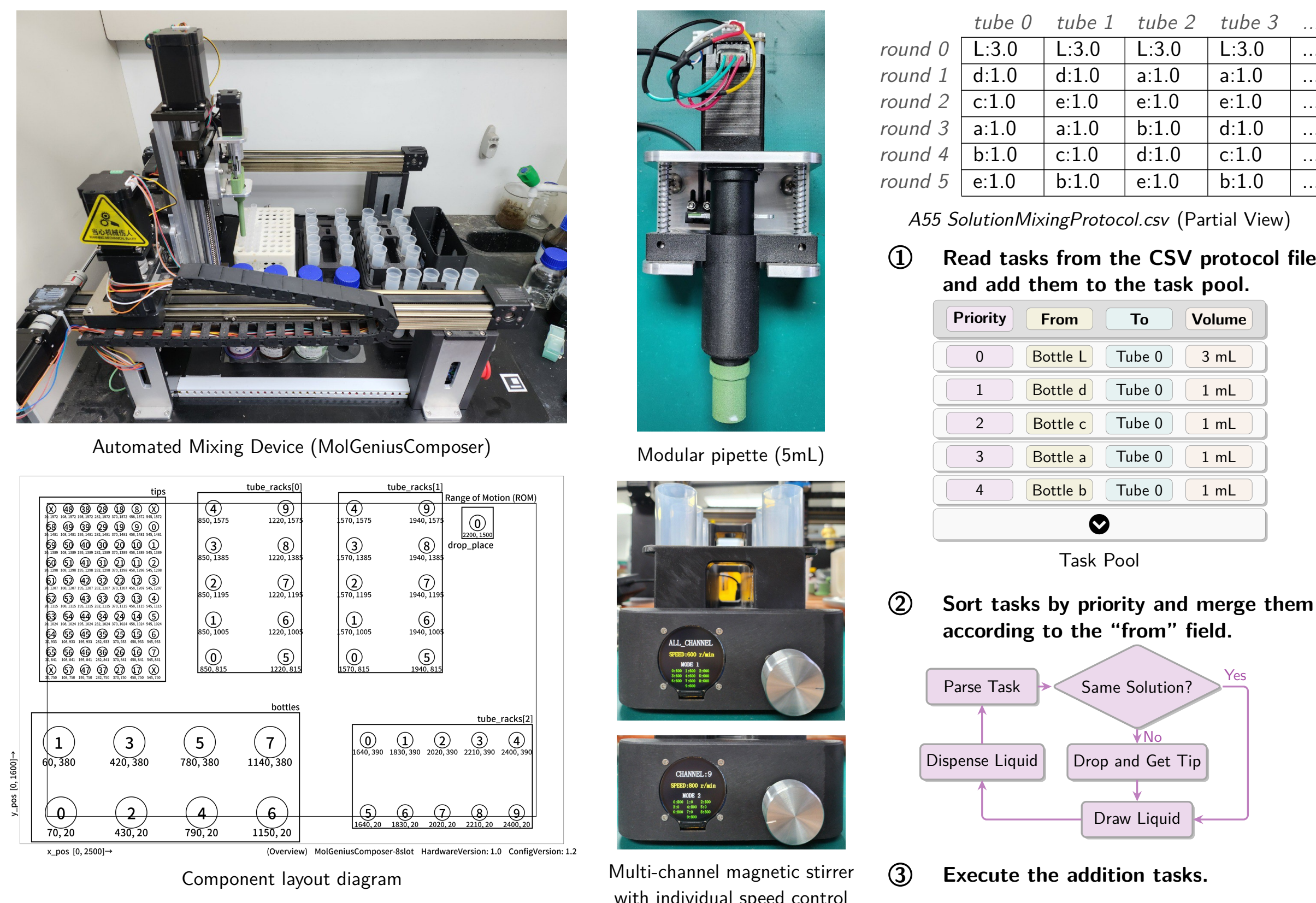


Figure 5. The Automated Mixing Device automated liquid handling and mixing device consists of an X-Y-Z motion module, a modular pipette, and a multi-channel magnetic stirrer. The device reads a CSV protocol file to manage tasks, sorts them by priority, merges based on the “from” field, and executes liquid addition. The stirrer provides individual speed control for each tube, ensuring proper mixing during the process. This system automates complex liquid transfers to improve efficiency and reproducibility.